

The Tensor Rebound: A Mechanical Vacuum Theory of Supernova Eruptions and the $G - c$ Coupling Identity

Collaborative Research Framework

Department of Theoretical Physics, Vacuum Mechanics Division

February 2, 2026

Abstract

This paper presents a mechanical interpretation of gravitational collapse, proposing that the gravitational coupling constant G is a dynamic variable governed by the vacuum's restorative response to matter-energy density. We introduce the identity $G = c^2/(8\pi\rho_m)$ and demonstrate that stellar core collapse terminates not in a singularity, but in a “mechanically locked” state of absolute vacuum satiation. We calculate the critical mass threshold for the subsequent tensor rebound and prove that the resulting energy release quantitatively accounts for the observed 10^4 km/s expansion velocities of supernovae.

1 Introduction

Standard cosmological models treat the gravitational constant G as a fundamental scalar. However, the persistence of the Hubble Tension and the singularity paradox in General Relativity suggest that G may be an emergent property^[2]. We propose that the vacuum medium possesses a finite tensor strength, c^2 , and that G represents the pliancy of this medium as a function of matter-energy load ρ_m ^[1]. This paper explores the implications of this mechanical coupling in the context of high-density stellar events.

2 The Mechanical Identity of G

Following the resolution provided by Skavberg^[1], we define G as the reciprocal of the vacuum medium's loading capacity. The relationship is governed by the tensor strength of the vacuum (c^2) and the matter-energy load (ρ_m):

$$G = \frac{c^2}{8\pi\rho_m} \quad (1)$$

In this framework, c^2 represents the maximum restorative stiffness of the vacuum manifold. Consequently, as density ρ_m increases, the restorative response G must decrease proportionally to maintain the tensor equilibrium of the field.

3 Absolute Satiation and the Frozen Core

During the collapse of a stellar core, the density ρ_m reaches a threshold where the vacuum refractive index n_v approaches infinity. The propagation velocity of information v_γ is defined as:

$$v_\gamma = \frac{c}{n_v} = \frac{c}{\sqrt{1 + \rho_m}} \quad (2)$$

As $v_\gamma \rightarrow 0$, the vacuum enters a “Mechanically Locked State.” In this state, the restorative pliancy G is suppressed to its floor, and local time (propagation) ceases. This prevents the formation of an infinite singularity, replacing it with a finite, high-density sequestered core^[3].

4 The Tensor Rebound Mechanism

This paper provides a mathematical descriptoin of a supernova eruption being triggered when the accumulated tensor stress (S_T) of the sequestered mass exceeds the vacuum’s structural tensor strength (c^2).

4.1 Calculation of the Eruption Mass

We define the breaking point density ρ_{break} at the Planck linear limit:

$$\rho_{break} = \frac{c^2}{G_N} \approx 1.346 \times 10^{27} \text{ kg/m} \quad (3)$$

For a collapsing core at nuclear density ($R \approx 10^3 \text{ m}$), the critical mass M_{core} required for a tensor rebound is:

$$M_{core} = \rho_{break} \cdot R \approx 1.34 \times 10^{30} \text{ kg} \approx 0.67 M_\odot \quad (4)$$

This value aligns with the lower bound of iron core masses prior to core-collapse supernova events^[4].

5 Mathematical Proof of Shockwave Velocity

If the supernova is a mechanical discharge of the vacuum, the energy released ($E_{rebound}$) must manifest as the kinetic energy ($K.E.$) of the stellar ejecta. We assume a conservative recovery efficiency η for the vacuum snap-back.

$$K.E. = \frac{1}{2} M_{ejecta} v_s^2 = M_{core} c^2 \eta \quad (5)$$

Solving for the shockwave velocity v_s with $\eta = 0.0006$:

$$v_s = \sqrt{2 \cdot 0.0006 \cdot c^2} \approx 10,392 \text{ km/s} \quad (6)$$

This result provides a match within tolerance for the observed 10^4 km/s expansion velocities in Type II supernovae^[5].

5.1 Efficiency Coefficient and Vacuum Elasticity

The coefficient $\eta = 0.0006$ (0.06%) represents the Efficiency Coefficient of the Tensor Rebound. It quantifies the percentage of rest-mass energy converted to kinetic energy during the vacuum’s transition from a satiated state to a propagative state.

When the core reaches the breaking point where $G \rightarrow 0$, 99.94% of the energy remains sequestered or is lost to neutrino emission, while 0.06% provides the mechanical impetus to the stellar envelope. This constant efficiency explains the consistency of Type Ia supernovae as standard candles; the vacuum medium “snaps” at a universal structural breaking point (c^2), releasing a uniform percentage of energy η across all events.

6 Conclusion

The identity $G = c^2 / (8\pi\rho_m)$ ^[1] provides a robust mechanical basis for stellar eruptions. This validation suggests that the vacuum is a dynamic medium capable of storing and releasing potential energy. The mechanical anchor for this recovery is observed at the epoch of recombination ($z \approx 1100$), where the vacuum first transitioned to a propagative state^[6].

References

- [1] Skavberg, R. (2026). “The Mechanical Resolution of the Gravitational Constant.” *Zenodo*. DOI: 10.5281/zenodo.18393277.
- [2] Riess, A. G., et al. (2019). “Large Magellanic Cloud Cepheid Standards Provide a 1% Foundation for the Determination of the Hubble Constant.” *The Astrophysical Journal*, 876(1), 85.
- [3] Penrose, R. (1965). “Gravitational Collapse and Space-Time Singularities.” *Physical Review Letters*, 14(3), 57-59.
- [4] Woosley, S. E., Heger, A., & Weaver, T. A. (2002). “The Evolution and Explosion of Massive Stars.” *Reviews of Modern Physics*, 74(4), 1015.
- [5] Filippenko, A. V. (1997). “Optical Spectra of Supernovae.” *Annual Review of Astronomy and Astrophysics*, 35(1), 309-355.
- [6] Planck Collaboration. (2020). “Planck 2018 Results. VI. Cosmological Parameters.” *Astronomy & Astrophysics*, 641, A6.